

Web Fundamentals



WEF1VO

Teil 4: CSS Box-Modell & Responsive Web

Medientechnik und -design | WS 2025/26

CSS Box-Modell

Berechnen der Gesamtgröße von Elementen

W3C Box- Model

```
width: 90px;  
height: 40px;  
padding: 10px;  
border: 5px;  
margin: 20px;
```

```
box-sizing:  
content-box;
```



Dimensionen



```
div {  
  width: 100px;  
  height: 50px;  
  /* min-width: 30px; */  
  /* min-height: 20px; */  
  /* max-width: 200px; */  
  /* max-height: 90px; */  
  overflow: auto;  
}
```

visible, hidden,
clip, scroll

Padding



```
div {
```

```
padding-top: 20px;
```

```
padding-right: 30px;
```

```
padding-bottom: 50px;
```

```
padding-left: 40px;
```

```
padding: 20px 30px 50px 40px;
```

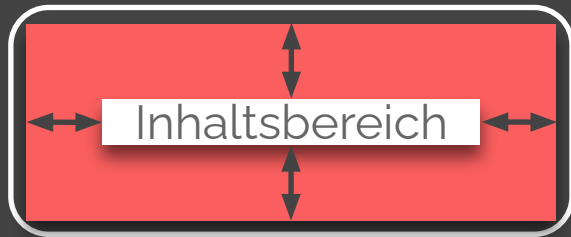
```
/* padding: 20px 35px 50px; */
```

```
/* padding: 35px 40px; */
```

```
/* padding: 50px; */
```

```
}
```

```
<div>
```



o r u l

o r l u

ou r l

or u l

Border



```
div {  
  border-width: 2px;  
  border-style: solid;  
  border-color: red;  
  border: 2px solid red;  
}
```

<div>



Margin



```
div {  
  margin-top: 20px;  
  margin-right: 30px;  
  margin-bottom: 50px;  
  margin-left: 40px;  
  margin: 20px 30px 50px 40px; */  
/* margin : 20px 35px 50px; */  
/* margin : 35px 40px; */  
/* margin : 50px; */  
}
```

o r u l
o r l u
ou r l
or u l

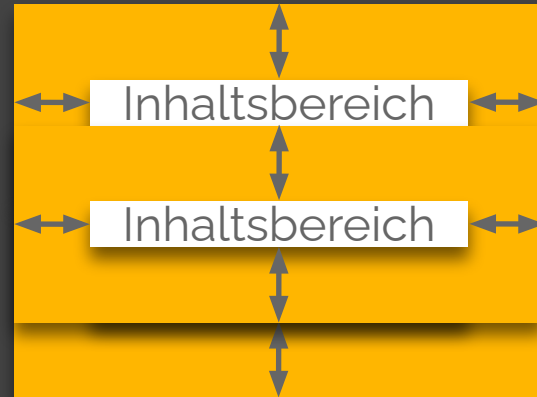


Margin-Collapse



```
<div>Inhaltsbereich</div>  
<div>Inhaltsbereich</div>
```

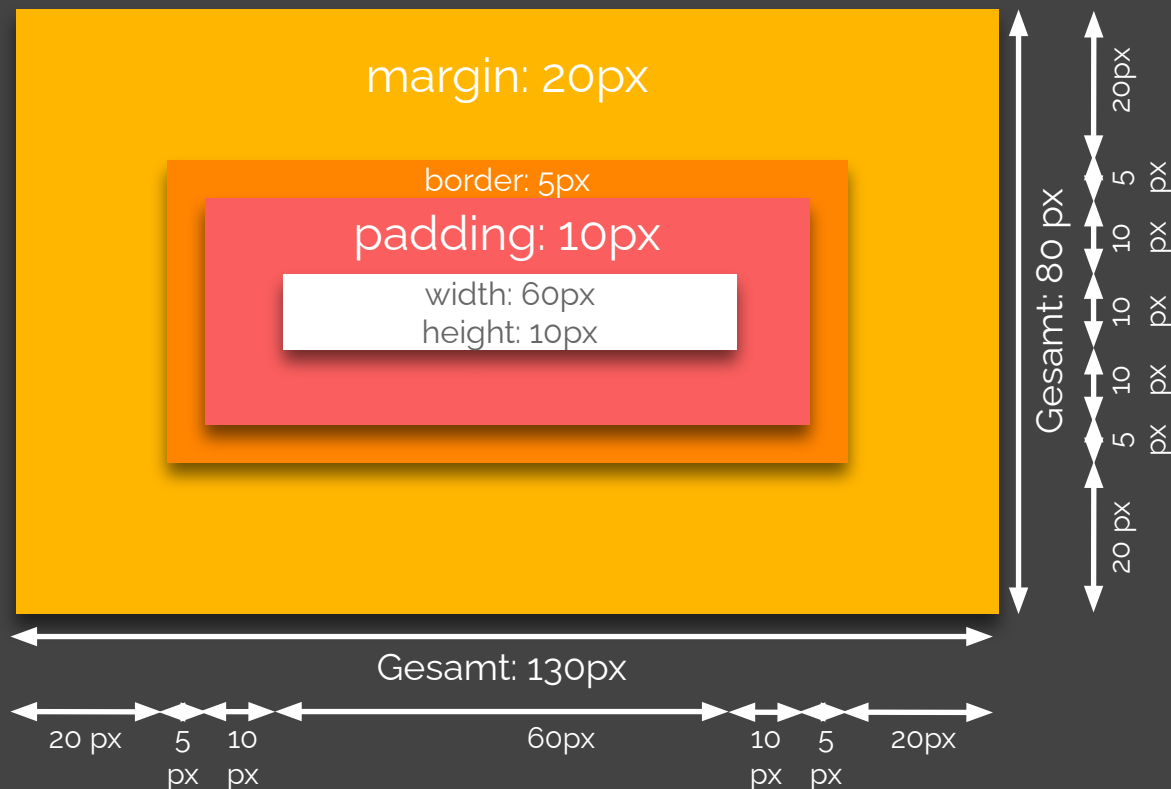
```
div {  
  margin: 20px;  
}
```



Alt. (IE) Box- Model

```
width: 90px;  
height: 40px;  
padding: 10px;  
border: 5px;  
margin: 20px;
```

```
box-sizing:  
border-box;
```





Die **Box** wechseln

Blockelemente: `<p>`, `<div>`, `<h1>`, `<header>`, ...

Inline-Elemente: `<em>`, `<img>`, `<a>`, `<strong>`, ...

```
em {  
    display: block;  
}
```

inline, block, inline-block, list-item, none (und mehr)

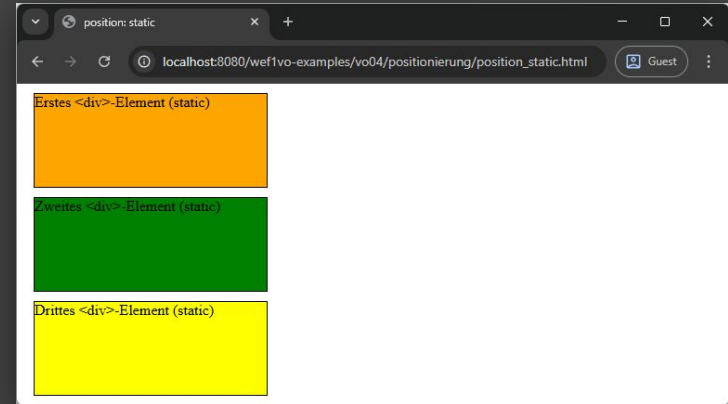
Positionierung

Elemente auf der Seite anordnen

Statische Positionierung



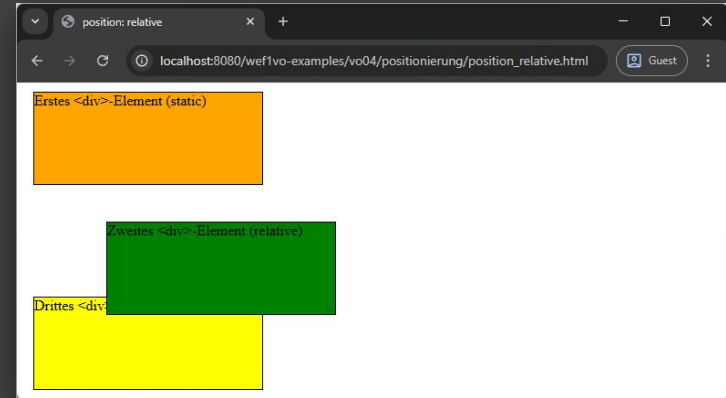
```
div {  
  border: 1px solid black;  
  width: 250px;  
  height: 100px;  
  margin: 10px;  
  position: static;  
}
```



Relative Positionierung



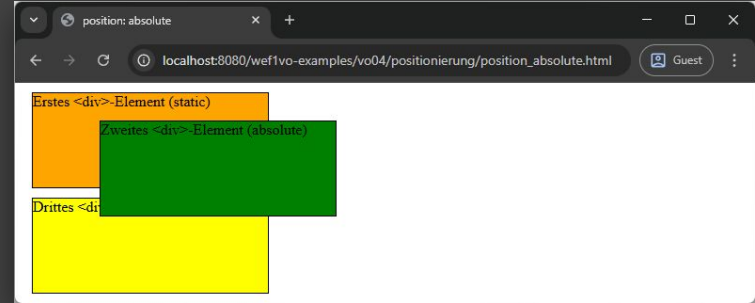
```
#two {  
  position: relative;  
  left: 80px;  
  top: 30px;  
  background-color: green;  
  /* z-index: 1; */  
}
```



Absolute Positionierung



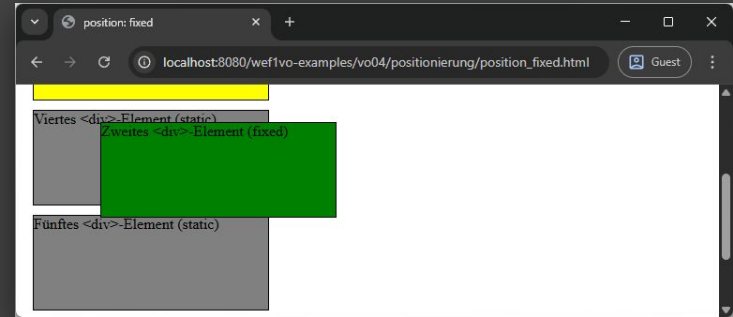
```
#two {  
  position: absolute;  
  left: 80px;  
  top: 30px;  
  background-color: green;  
  /* z-index: 1; */  
}
```



Fixe Positionierung



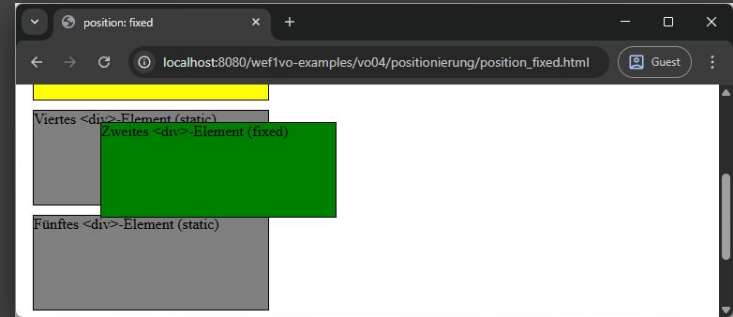
```
#two {  
  position: fixed;  
  left: 80px;  
  top: 30px;  
  background-color: green;  
  /* z-index: 1; */  
}
```



Sticky Positionierung



```
#two {  
  position: sticky;  
  left: 80px;  
  top: 30px;  
  background-color: green;  
  /* z-index: 1; */  
}
```



Elemente floaten

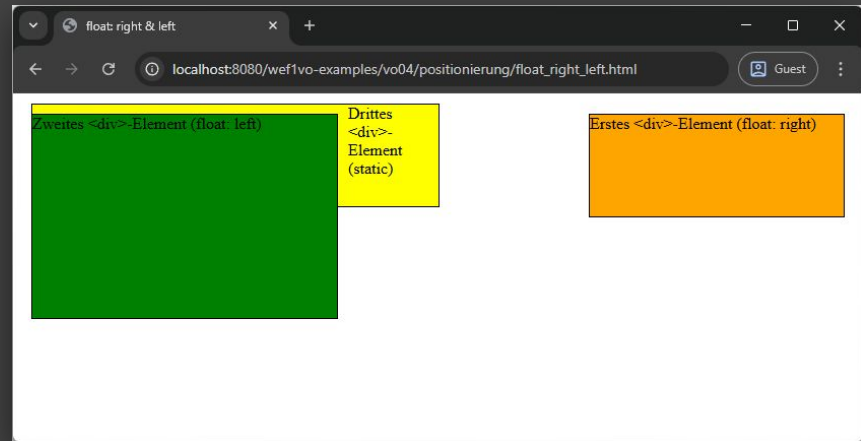
Elemente aus dem Document Flow nehmen
und sie links oder rechts verschieben.



float: left, float: right



```
#one {  
  background-color: orange;  
  float: right;  
  width: 200px;  
  height: 100px;  
}  
#two {  
  background-color: green;  
  float: left;  
  width: 300px;  
  height: 200px;  
}
```



clear: **left**, clear: **right**



```
#three {  
  width: 400px;  
  height: 100px;  
  clear: left;  
}
```





Anordnen mit

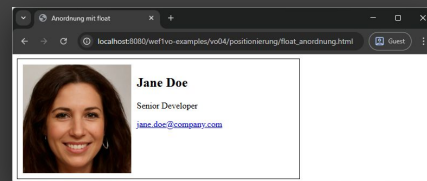
float

```
<section class="staff-card">
  <div class="staff-photo">
    
  </div>
  <div class="staff-details">
    <h2>Jane Doe</h2>
    <p>Senior Developer</p>
    <p>
      <a href="mailto:jane.doe@company.com">
        jane.doe@company.com</a>
      </p>
    </div>
</section>
```

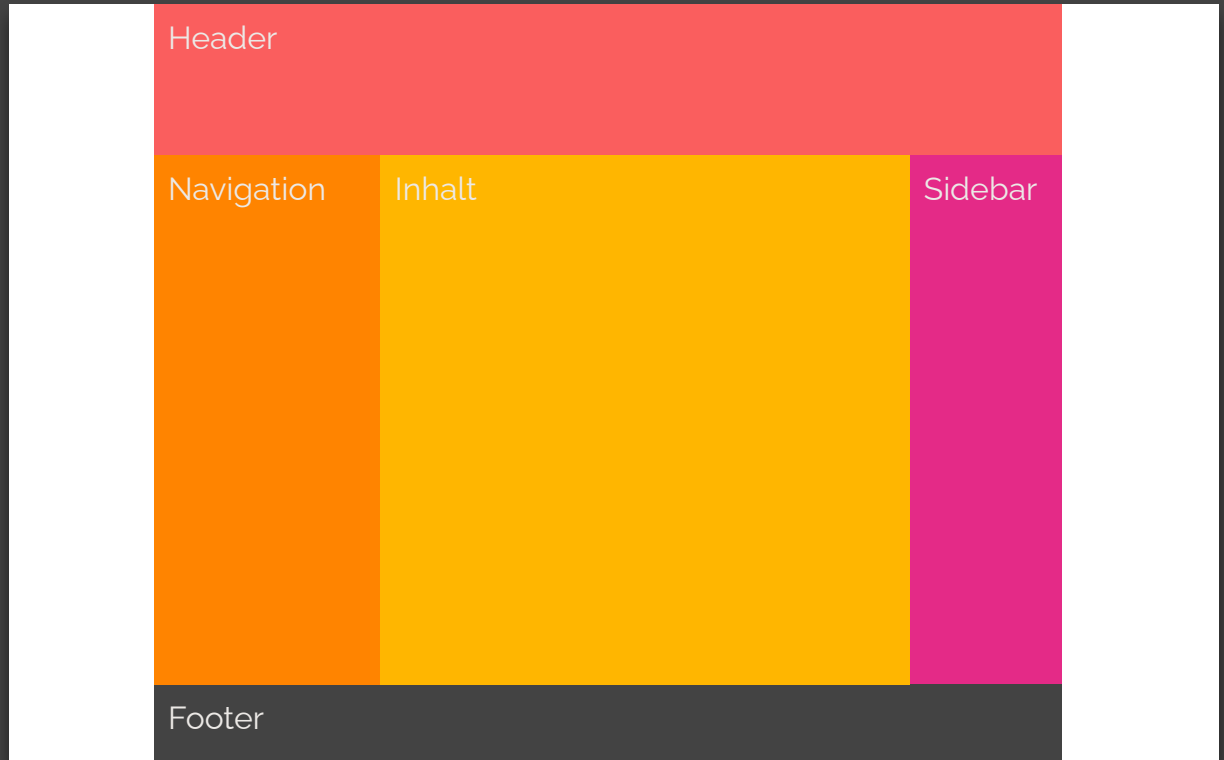
```
.staff-card {
  width: 500px;
  height: 200px;
  padding: 10px;
  border: 1px solid black;
}

.staff-photo {
  height: 100%;
  float: left;
  margin-right: 10px;
}

.staff-photo img {
  height: 100%;
}
```



Layout s mit float



Das Markup



```
<body>
  <div id="container">
    <header>
      <h1>Überschrift</h1>
    </header>
    <nav>
      <ul>
        <li>Navigation 1</li>
        <li>Navigation 2</li>
      </ul>
    </nav>
    <main>
      <section id="content">
        <p>Lorem ipsum dolor sit amet, [...]</p>
      </section>
      <aside>
        <p>Lorem ipsum dolor sit amet, [...]</p>
      </aside>
    </main>
    <footer>
      <p>&copy; 2025 Der*Die Autor*in</p>
    </footer>
  </div>
</body>
```

Die Styles



```
* {
  margin: 0;
  box-sizing: border-box;
}

body {
  color: rgb(236 233 230);
  font-family: Raleway, Arial, sans-serif;
  line-height: 2em;
}

div#container {
  width: 960px;
  margin: 0 auto;
  background-color: rgb(255 132 0);
}

header {
  background-color: rgb(250 94 94);
}

nav {
  width: 240px;
  float: left;
}
```

```
main {
  width: 720px;
  float: right;
  background-color: rgb(228 42 135);
}

section#content {
  width: 600px;
  float: left;
  background-color: rgb(255 182 0);
  color: rgb(67 67 67);
}

aside {
  width: 120px;
  float: right;
}

footer {
  clear: both;
  background-color: rgb(67 67 67);
}

header, nav, footer, section, aside {
  padding: 0.5em;
}
```

Media Queries

Webseiten an Ausgabegeräte anpassen



Medienspezifische Stylesheets

Via <link>-Element:

```
<link rel="stylesheet"
      href="style.css"
      media="screen">
```

```
<link rel="stylesheet"
      href="print.css"
      media="print">
```

Via @import:

```
@import
  url("style.css")
  screen;
```

```
@import
  url("print.css")
  print;
```

Direkt in CSS:

```
@media print {
  nav {
    display: none;
  }
}
```



Medientypen

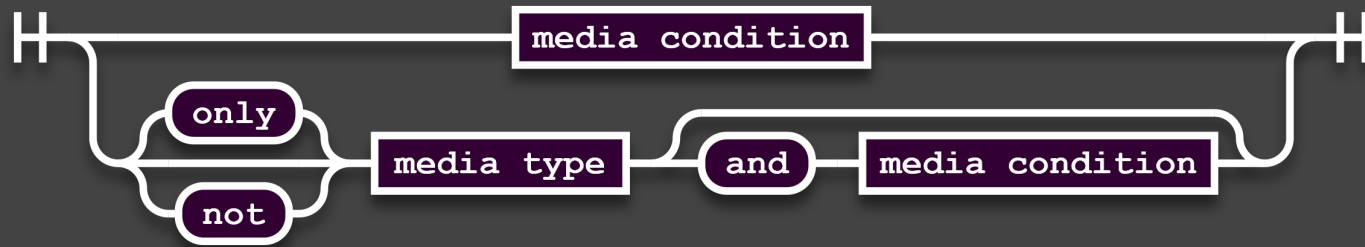
Verwendbar

<code>all</code>	Alle Ausgabegeräte
<code>print</code>	Drucker und Druckvorschau
<code>screen</code>	Bildschirmbasierte Geräte (auch Screenreader)

Deprecated (veraltet)

<code>tty</code>	(textbasierte Computerterminals)
<code>tv</code>	(TV-Geräte)
<code>projection</code>	(Projektoren)
<code>handheld</code>	(mobile Geräte mit geringer Auflösung)
<code>braille</code>	(Braille-Zeilen-basierte Ausgabegeräte),
<code>embossed</code>	(Braille-Drucker),
<code>aural</code>	(Sprachsynthesizer) und <code>speech</code>
	(Screenreader).

Aufbau einer Media Query



```
@media only screen and (max-width: 480px) {  
  /* CSS-Regeln */  
}
```



Merkmale verknüpfen

Logisches Und:

```
@media screen and  
  (min-width: 768px) and  
  (max-width: 960px) {  
  /* CSS-Regeln */  
}
```

Logisches Oder:

```
@media  
  (min-height: 680px),  
  screen and  
  (orientation: portrait) {  
  /* CSS-Regeln */  
}
```

Logische Verneinung:

```
@media not (color) {  
  /* CSS-Regeln */  
}
```



Medienmerkmale

`width` (min-width, max-width)
`height` (min-height, max-height)
`orientation` [portrait, landscape]
`aspect-ratio` (min-aspect-ratio, max-aspect-ratio)
`color` (min-color, max-color)
`resolution` (min-resolution, max-resolution)
`pointer` [none, coarse, fine]
`prefers-reduced-motion` [no-preference, reduce]
`prefers-contrast` [no-preference, high, low, forced]



Immer Viewport!

Viewport-Angabe im HTML-Dokument:

```
<head>  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  ...  
</head>
```

Zusätzliche Angaben:

```
user-scalable=no  
Minimum-scale=1.0  
maximum-scale=2.0
```



Responsive Layout

```
<body>
  <div id="container">
    <header>
      <h1>Überschrift</h1>
    </header>
    <nav>
      <ul>
        <li>Navigation 1</li>
        <li>Navigation 2</li>
      </ul>
    </nav>
    <main>
      <section id="content">
        <p>Lorem ipsum dolor sit amet, [...]</p>
      </section>
      <aside>
        <p>Lorem ipsum dolor sit amet, [...]</p>
      </aside>
    </main>
    <footer>
      <p>&copy; 2025 Der*Die Autor*in</p>
    </footer>
  </div>
</body>
```



Desktop first



```
div#container {  
  width: 960px;  
  margin: 0 auto;  
  background-color: rgb(255 132 0);  
}  
  
nav {  
  width: 25%;  
  float: left;  
}  
  
main {  
  width: 75%;  
  float: right;  
  background-color: rgb(228 42 135);  
}  
  
section#content {  
  width: 80%;  
  float: left;  
  background-color: rgb(255 182 0);  
  color: rgb(67 67 67);  
}  
  
aside {  
  width: 20%;  
  float: right;  
}
```

```
@media screen and (max-width: 960px) {  
  div#container {  
    width: 100%;  
  }  
}
```

```
@media screen and (max-width: 768px) {  
  nav {  
    float: none;  
    width: 100%;  
    text-align: center;  
  }  
  
  main {  
    width: 100%;  
  }  
}
```

```
@media screen and (max-width: 480px) {  
  section#content, aside, main {  
    float: none;  
    width: 100%;  
  }  
}
```


Mobile first

```
div#container {  
  background-color: rgb(255 132 0);  
}  
  
nav {  
  text-align: center;  
}  
  
nav ul {  
  padding: 0;  
}  
  
nav ul li {  
  list-style-type: none;  
  display: inline;  
}  
  
nav ul li:not(:last-child) {  
  margin-right: 0.5em;  
}  
  
main {  
  background-color: rgb(228 42 135);  
}  
  
section#content {  
  background-color: rgb(255 182 0);  
}
```

```
@media screen and (min-width: 480px) {  
  main {  
    width: 100%; float: right;  
  }  
  section#main {  
    float: left; width: 80%;  
  }  
  section#sidebar {  
    float: left; width: 20%;  
  }  
  footer {  
    clear: both;  
  }  
}  
  
@media screen and (min-width: 768px) {  
  nav {  
    float: left; width: 25%;  
    text-align: left;  
  }  
  main {  
    width: 75%;  
  }  
}  
  
@media screen and (min-width: 960px) {  
  div#container {  
    width: 960px; margin: 0 auto;  
  }  
}
```





Nächstes Mal in WEF1VO

Layout-Frameworks

Credits



- S. 27: Media Query Syntax Scheme: <https://www.w3.org/TR/mediaqueries-5/>